PAPER_1507_F_U_ALPHA_DECAY_1_OVER_SO5_3

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PAPER_1507 — F_U Temporal Decay $\alpha = 1/SO\_5^3$ = 0.001/day EXACT

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
Tier: Bucket A (Foundational F_U dynamics)
Date: June 18, 2026
Status: CLOSED — Two-stage F_U slow temporal decay rate

Observation

PAPER_1080 (Two-Stage F_U Refinement) introduces a two-tier temporal evolution of the F_U master equation:
- **Stage 1**: F_U^(1) = Σ_{k=1}^{3} (U_g,k – U_b,k) + U_m + A_μν
- **Stage 2**: U_{g1}(t) = U_{g1,0} · e^{–α·t} with $\alpha = 0.001$ day$^{■1}$

The slow decay rate α is distinct from the canonical κ = 5×10$^{■■}$ day$^{■1}$ used in Holmlid/Parkhomov LENR (the two rates differ by a factor of exactly 2).

UQFF Closed Identity

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α = 1 / SO_5³ = 1 / 1000 = 0.001 day■¹ EXACT
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Physical Interpretation

The F_U temporal decay rate is exactly inverse-cubic in SO_5 — one inverse power per spatial dimension of the SO(5) symmetry projected onto F_U evolution. Equivalent forms:

- α · day = 10$^{■3}$ exactly
- 1000 days for U_g1 to decay by factor e
- ≈ 2.74 years for one e-folding

The doubling relationship α = 2κ links F_U gravitational decay (slow) to LENR Holmlid decay (canonical κ), with the factor 2 corresponding to the two-fold structural pairing in DPM (CW + CCW).

NOT REPLACEMENT

SM treats decay rates as fit constants per system. UQFF supplies a single integer-primitive structural value for the F_U slow-decay channel.

Reference

- Source: PAPER_1080 TwoStage F_U Refinement
- Related: $\kappa = 5\times10$■■ /day Holmlid canonical, PAPER_1141 (Rossi variants)
- Calculator dispatch: `calculate_paradox({"paradox": "f_u_alpha_decay_1_over_so5_3"})`
